Week 14 Homework

Following Section 2.3 of *Discrete Mathematics, an Open Introduction*. Let $\{a_n\}$ be a sequence of real numbers. The k^{th} **forward difference** of this sequence are defined recursively:
The **first forward difference** Δa_n is an new sequence defined by:$\Delta a_n = a_{n+1} - a_n$.
The k**th forward difference** $\Delta^k a_n$ is $\Delta^k a_n = \Delta^{k-1} a_{n+1} - \Delta^{k-1} a_n$. So if the sequence for a_n is $1, 3, 5, 7, 9, \ldots$ then Δa_n is $3-1, 5-3, 7-5, 9-7, \ldots = 2, 2, 2, 2, 2, \ldots$.

Exercise

Find Δ and Δ^2 for the following sequences :
a. $2, 4, 6, 8, 10, 12, 14, \ldots$
b. $1, 4, 9, 16, 25, 36, 49, \ldots$

Exercise

Make up sequences that have:
a. 3, 3, 3, 3, … as its second differences.
b. 1, 2, 3, 4, 5, … as its third differences.
c. 1, 2, 4, 8, 16, … as its fifth differences.

Exercise

Find the closed form solution to the sequence using the using the polynomial fitting method for the sequence $0, 3, 8, 15, 24, 35, 48, 63, 80, \ldots$ which has the recurrence $a_n = a_{n-1} + 2n + 1$ with $a_0 = 0$.

Exercise

Find the closed form solution to the sequence using the using the polynomial fitting method for the sequence $1, 4, 10, 20, 35, 56, 84, \ldots$ which has the recurrence $a_n = 4a_{n-1} - 6a_{n-2} + 4a_{n-3} - a_{n-4}$ with $a_1 = 1, a_2 = 4, a_3 = 10$ and $a_4 = 20$. This also happens to the be the sequence given by summing up the triangular numbers. That is, $a_n = \sum_{i=1}^{n} T_n$.

Exercise

Calculate $\Delta\binom{n}{4}$. Use this to rewrite $\Delta\binom{n}{k}$ as a binomial coefficient.

Proof

Prove $\Delta(\Delta^i a_n) = \Delta^{i+1} a_n$.

Proof

Let $b_n = n^3$. Prove $\Delta^2 b_n = 6n + 6$.

Proof

Let $b_n = n^k$. Prove $\Delta b_n = \sum_{i=0}^{k-1} \binom{k}{i} n^i$

Proof

Let $a_n = \alpha^n$ for $n \geq 0$ and α a real number. Prove $\Delta^k a_n = (\frac{\alpha-1}{\alpha})^k a_n$ for all $k \geq 0$.

Proof

Let a_n and b_n be sequences. Prove $\Delta(a_n \cdot b_n) = a_{n+1}\Delta b_n + b_n \Delta a_n$.